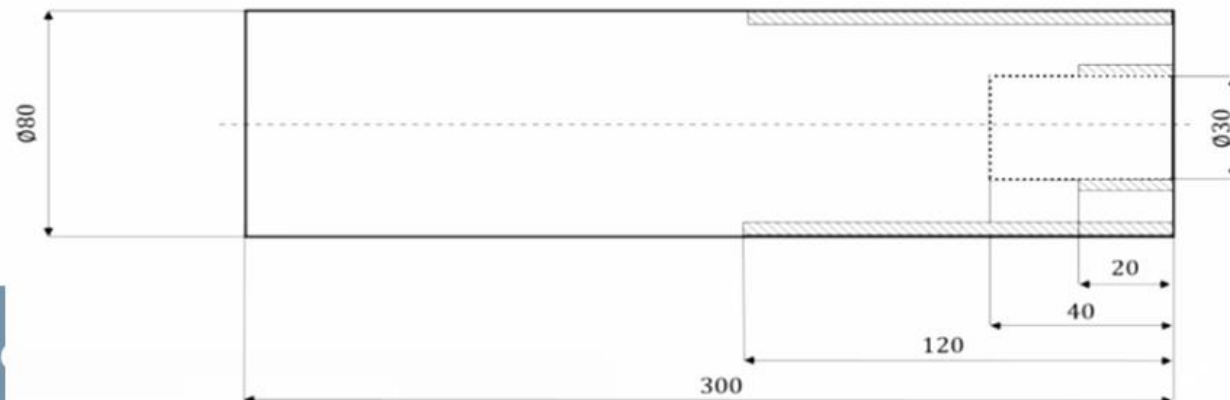


■ Problem 5 (12/07/2021)

The mechanical component in the figure must be machined on a lathe. The machining operations are an external cylindrical turning of a length equal to 120 mm and an internal cylindrical turning of a length equal to 20 mm. A metal bar of 80 mm diameter is mounted on a self-centering 3-jaw chuck and machined according to the following parameters (the two machining operations are carried out with tools with identical characteristics):

- External cylindrical turning: depth of cut = 1 mm, feed = 0,4 mm/rev.
 - Internal cylindrical turning: depth of cut = 0,5 mm, feed = 0,1 mm/rev.
 - Other data: Cutting pressure: $k_{c0,4} = 2600$ MPa, Kronenberg coefficient $x = 0,29$, Main entering angle = 95° , Secondary entering angle = 10° , Tool nose radius = 0,4 mm, Nominal lathe power = 5 kW, Lathe efficiency = 90%.
1. Determine the cutting force, the cutting power, and the cutting time for the external cylindrical turning considering a cutting speed is 250 m/min and determine if the machining operation is feasible.
 2. Determine the average roughness on the machined surface of the hole.



MACHINING : TURNING OPERATIONS

▪ **Solution - Q1**

Determine the cutting force, the cutting power, and the cutting time for the external cylindrical turning considering a cutting speed is 250 m/min and determine if the machining operation is feasible.

Using the cutting pressure method, the cutting force is equal to $F_C = k_c A_D$, where

$$k_c = k_{c,0,4} \left(\frac{0,4}{f \cdot \sin k_r} \right)^x = 2600 \cdot \left(\frac{0,4}{0,4 \cdot \sin 95^\circ} \right)^{0,29} = 2.603 \text{ MPa}$$
$$A_D = f \cdot a_p = 0,4 \cdot 1 = 0,4 \text{ mm}^2$$

Therefore

$$F_C = 2.603 \cdot 0,4 = 1041 \text{ N}$$

The cutting power is

$$P_C = v_c \cdot F_C = \frac{250 \cdot 1041}{60} = 4.338 \text{ W}$$

The constraint on the cutting power is

$$P_{limit} = P_g \cdot E = 5 \cdot 0,9 = 4,5 \text{ kW}$$



MACHINING : TURNING OPERATIONS

▪ Solution - Q1

Determine the cutting force, the cutting power, and the cutting time for the external cylindrical turning considering a cutting speed is 250 m/min and determine if the machining operation is feasible.

Being $P_C \leq P_{limit}$, the external cylindrical turning is feasible.

The cutting time is equal to $t_c = \frac{l_c}{v_f}$, where, from the figure,

the cutting length is $l_c = 120 + 1 = 121$ mm

the final diameter is $D_f = D_0 - 2 \cdot a_p = 80 - 2 \cdot 1 = 78$ mm

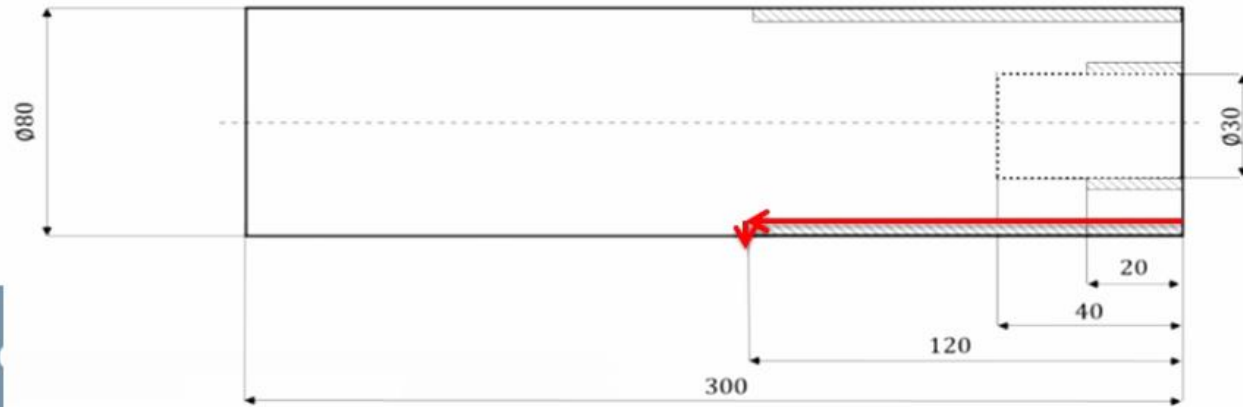
the rotational speed is

$$n = \frac{1000 \cdot v_c}{\pi \cdot D_f} = \frac{1000 \cdot 250}{\pi \cdot 78} = 1020 \text{ RPM}$$

and the feed velocity is $v_f = f \cdot n = 0,4 \cdot 1020 = 408$ mm/min.

Therefore,

$$t_c = \frac{l_c}{v_f} = \frac{121}{408} \cdot 60 = 17,8 \text{ s}$$



MACHINING : TURNING OPERATIONS

▪ **Solution - Q2**

Determine the average roughness on the machined surface of the hole.

Using the Schmalz equation, we get

$$R_a = \frac{1}{32} \cdot \frac{f^2}{r} \cdot 10^3 = \frac{1}{32} \cdot \frac{0,1^2}{0,4} \cdot 10^3 = 0,8 \mu\text{m}$$

To apply this equation, the following constraints must be verified:

$$\begin{aligned} f &= 0,1 < 2 \cdot r \cdot \sin(k_r) = 2 \cdot 0,4 \cdot \sin(95^\circ) = 0,8 \text{ mm/rev.} \\ f &= 0,1 < 2 \cdot r \cdot \sin(k'_r) = 2 \cdot 0,4 \cdot \sin(10^\circ) = 0,14 \text{ mm/rev.} \end{aligned}$$

The maximum feed limit is $f_{limit} = 0,14 \text{ mm/rev}$, so the equation can be applied.

